Huaiyu Wei

Los Angeles, CA | hywei@atmos.ucla.edu | (310) 696-3887

huaiyuwei.github.io | Google Scholar | ResearchGate | GitHub

Employment

University of California, Los Angeles (UCLA) Feb. 2024 – Present
Postdoctoral Researcher
Mentors: Andrew L. Stewart & James C. McWilliams

Education

The Hong Kong University of Science and Technology (HKUST) Aug. 2019 – Feb. 2024
Ph.D. in Marine Environmental Science, Advisor: Yan Wang
Thesis: Parameterizing mesoscale eddy fluxes across continental slopes

Sun Yat-sen University (SYSU) Sep. 2015 – July 2019
B.S. in Physical Oceanography, Advisor: Zhan Hu
Thesis: Laboratory study on wave dissipation by vegetation in combined current–wave flow

Boston University May 2023 – Aug. 2023
Visiting student, Advisor: Xiaozhou Ruan

University of British Columbia July 2017 – Aug. 2017
Summer Program on Ocean and Atmosphere Systems

Research Interests

Ocean circulation and climate; Meridional overturning circulation; Mesoscale eddies and eddy–topography interaction; Abyssal circulation and bottom boundary layer mixing; Eddy parameterization; Machine learning

Preprints

1. **Wei, H.**, Stewart, A. L., Medvedev, A., Srinivasan, K., Solodoch, A., Manucharyan, G., & Hogg, A. (2026). Deep learning unlocks satellite-based monitoring of the global ocean overturning circulation. *Under review*. Preprint: doi.org/10.22541/essoar.15004679/v1.
2. Medvedev, A., **Wei, H.**, Armour, K., Srinivasan, K., Solodoch, A., Stewart, A. L., & Manucharyan, G. (2026). Basin-wide sea-surface observations reveal post-2000 emergence of AMOC weakening. *Under review*. Preprint: doi.org/10.22541/essoar.15002298/v1.

Publications

1. **Wei, H.**, Stewart, A. L., McWilliams, J. C., & Capó, E. (2026). Formation of abyssal downwelling-favorable prograde flows via mesoscale eddy potential vorticity mixing. *Journal of Physical Oceanography*, 56(6), 1363–1385. doi.org/10.1175/JPO-D-25-0196.1.
2. **Wei, H.**, Srinivasan, K., Stewart, A. L., Solodoch, A., Manucharyan, G., & Hogg, A. (2025). Full-depth reconstruction of long-term meridional overturning circulation variability from satellite-measurable quantities via machine learning. *Journal of Advances in Modeling Earth Systems*, 17(7), e2024MS004915. doi.org/10.1029/2024MS004915.
3. Li, D., **Wei, H.**, & Ruan, X. (2025). The importance of eddy stirring in wind-driven coastal upwelling. *Journal of Physical Oceanography*, 55(1), 29–42. doi.org/10.1175/JPO-D-24-0066.1.
4. **Wei, H.**, Wang, Y., & Mak, J. (2024). Parameterizing eddy buoyancy fluxes across prograde shelf/slope fronts using a slope-aware GEOMETRIC closure. *Journal of Physical Oceanography*, 54(2), 359–377. doi.org/10.1175/JPO-D-23-0152.1.
5. Xie, C.*, **Wei, H.***, & Wang, Y. (2023). Bathymetry-aware mesoscale eddy parameterizations across upwelling slope fronts: A machine learning-augmented approach. *Journal of Physical Oceanography*, 53(12), 2861–2891. doi.org/10.1175/JPO-D-23-0017.1. **(*Contributed equally)**

6. Xie, C., **Wei, H.**, & Wang, Y. (2023). Impact of parameterized isopycnal diffusivity on shelf-ocean exchanges under upwelling-favorable winds: offline tracer simulations augmented by artificial neural network. *Journal of Advances in Modeling Earth Systems*, 15(4), e2022MS003424. doi.org/10.1029/2022MS003424.
7. **Wei, H.**, Wang, Y., Stewart, A. L., & Mak, J. (2022). Scalings for eddy buoyancy fluxes across prograde shelf/slope fronts. *Journal of Advances in Modeling Earth Systems*, 14(12), e2022MS003229. doi.org/10.1029/2022MS003229. **(Issue cover)**
8. Hu, Z., Lian, S., Zitman, T., Wang, H., He, Z., **Wei, H.**, et al. (2022). Wave breaking induced by opposing currents in submerged vegetation canopies. *Water Resources Research*, 58(4), e2021WR031121. doi.org/10.1029/2021WR031121.
9. **Wei, H.**, & Wang, Y. (2021). Full-depth scalings for isopycnal eddy mixing across continental slopes under upwelling-favorable winds. *Journal of Advances in Modeling Earth Systems*, 13(6), e2021MS002498. doi.org/10.1029/2021MS002498. **(Issue cover)**
10. Hu, Z., Lian, S., **Wei, H.**, Li, Y., Stive, M., & Suzuki, T. (2021). Laboratory data on wave propagation through vegetation with following and opposing currents. *Earth System Science Data*, 13(10), 4987–4999. doi.org/10.5194/essd-13-4987-2021.

Research Support & Proposals

"Mesoscale eddy-mediated diapycnal upwelling in the ocean's sloping bottom boundary layers" NSF proposal (Physical Oceanography Program) — Pending
PI: Huaiyu Wei; Co-PIs: Andrew L. Stewart, James C. McWilliams, Henri Drake.

"Leveraging machine learning and satellite measurements to predict ocean meridional overturning circulation" NSF ACCESS Explore project (EES240055). — 2024–2026

Talks & Seminars

1. "Formation of abyssal prograde flows via eddy potential vorticity mixing", UC Irvine, 29 Jul. 2025.
2. "Reconstructing meridional overturning circulation from satellite measurements via neural networks", Boston University (20 Dec. 2024), MIT (3 Jan. 2025), UCLA AOS Department Seminar (19 Feb. 2025), and SNAP Seminar (24 Feb. 2025).
3. "Parameterizing eddy mixing across continental slopes under upwelling-favorable winds", Marine Center Spring Meeting, UCLA, 9 May 2024.
4. "Parameterizing isopycnal eddy mixing across continental slopes", AOS Ocean Seminar, UCLA, 9 Apr. 2024.

Conference Presentations & Workshops

California GFD Meeting 2026, UCLA, Los Angeles, CA, USA — Sept. 2026
Poster presentation: "Turning ocean mixing and residual overturning circulation upside down"

Ocean Sciences Meeting 2026, Glasgow, UK — Feb. 2026
Oral presentation: "Mesoscale eddy potential vorticity mixing drives abyssal prograde along-isobath circulation"

NCAR Regional Ocean Modeling Workshop, Boulder, CO, USA — Oct. 2025
Participant: Regional ocean modeling with MOM6 in the Community Earth System Model framework

California GFD Meeting 2025, UCSB, Santa Barbara, CA, USA — Sept. 2025
Oral presentation: "Eddy potential vorticity mixing drives abyssal prograde along-isobath circulation"

Climate Process Team Meeting 2025, NYU, New York, NY, USA (hybrid) — Aug. 2025
Oral presentation: "Formation of abyssal prograde flows via mesoscale eddy potential vorticity mixing"

AGU Fall Meeting 2024, Washington, DC, USA — Dec. 2024
Oral presentation: "Reconstructing meridional overturning circulation from satellite measurements via neural networks"

Ocean Sciences Meeting 2024, New Orleans, LA, USA — Feb. 2024
eLightning presentation: "Parameterizing eddy buoyancy fluxes across prograde shelf/slope fronts using a slope-aware GEOMETRIC closure"

Ocean Transport and Eddy Energy Meeting 2023, WHOI, Woods Hole, MA, USA — May 2023
Oral presentation: "Parameterization for eddy buoyancy fluxes across prograde shelf/slope fronts"

Ocean Sciences Meeting 2022, Online — Mar. 2022
Oral presentation: "Full-depth scalings for isopycnal eddy mixing across continental slopes under upwelling-favorable winds"

Selected Awards

Richard P. and Linda S. Turco Exceptional Research Publication Award, UCLA	2026
Nominee, Chancellor's Award for Postdoctoral Research, UCLA	2025, 2026
RedBird Academic Excellence Award, HKUST	2023
Best Presentation Award, HKUST Postgraduate Seminar	2023
Outstanding Graduate, SYSU	2019
National Scholarship, China (Top 1%)	2018
The Giordano Scholarship, SYSU (Top 2%)	2017
The Coca-Cola Scholarship for Outstanding Students, SYSU (Top 5%)	2016

Teaching Experience

Guest lecture, AOS 201A Geophysical Fluid Dynamics (graduate), UCLA	2025
Guest lectures (2), AOS 103 Physical Oceanography (undergraduate), UCLA	2024, 2025
Teaching assistant, HKUST: Survey of Ocean Science; Descriptive Physical Oceanography; The Earth as a Blue Planet	2020–2023

Research Mentoring

Undergraduate researchers at UCLA:

- Reid Mendoza (Mathematics): Machine-learning reconstruction of ocean meridional heat transport
- Evelyn Zhu (co-advised; Mathematics): Machine-learning reconstruction of top-of-atmosphere radiation from surface temperature in climate models
- Selena Yu (now Ph.D. student, Stanford): Antarctic meltwater and global overturning circulation
- Alex Akin (co-advised; now M.S. student, Moss Landing Marine Labs): Antarctic shelf eddy heat transport

Professional Service

Organizing Committee, California Geophysical Fluid Dynamics Meeting (UCLA)	2026
Organizing Committee, California Geophysical Fluid Dynamics Meeting (UCSB)	2025

Journal reviewer: Communications Earth & Environment, Deep-Sea Research, Geophysical Research Letters, Journal of Advances in Modeling Earth Systems, Journal of Geophysical Research: Oceans, Journal of Physical Oceanography, Nature Communications, Ocean Modelling.

Additional Information

Languages: Mandarin (native), English (fluent).
Numerical models: MITgcm, ROMS. **Programming:** Fortran, MATLAB, Python.